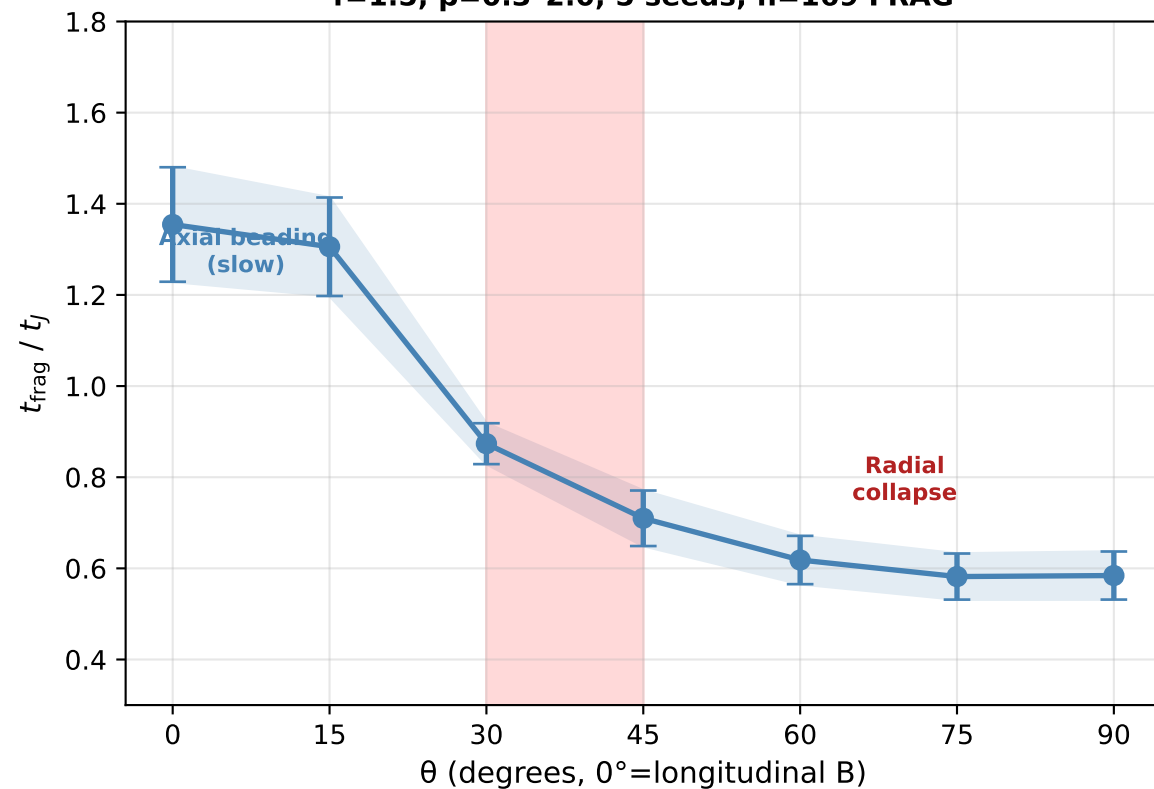
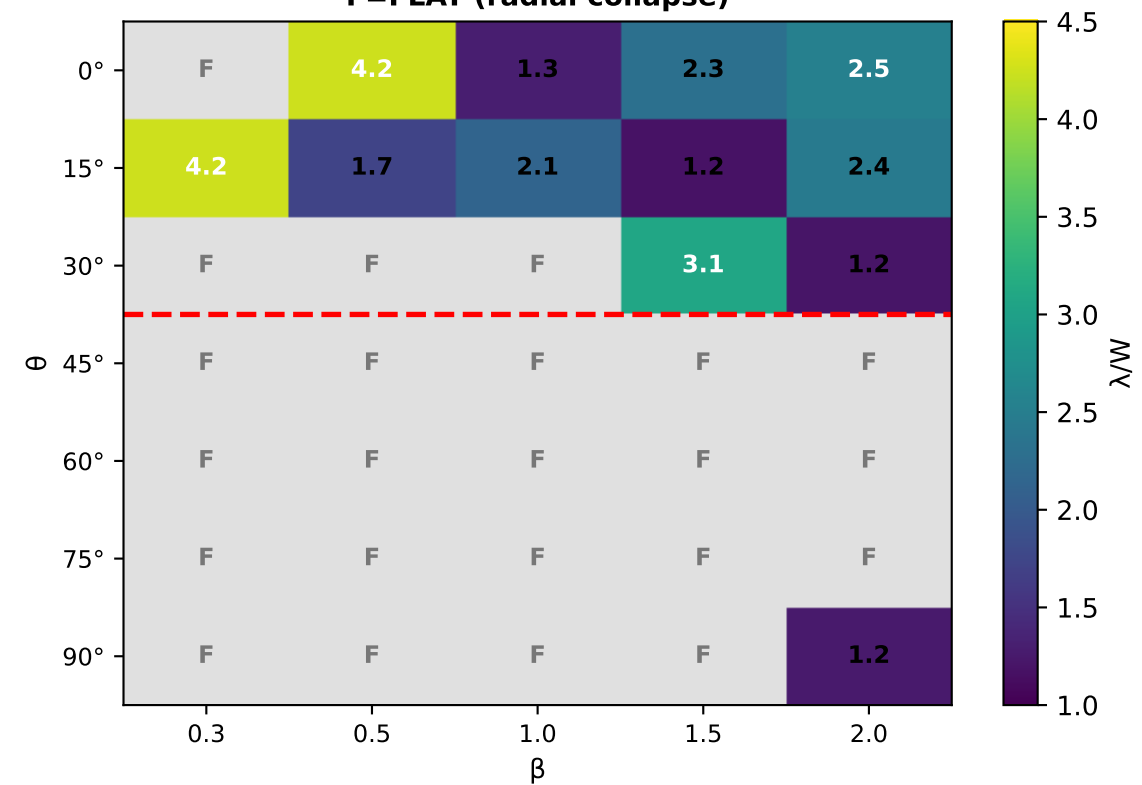


ASTRA Referee Response Campaigns C8-C12: Key Scientific Results

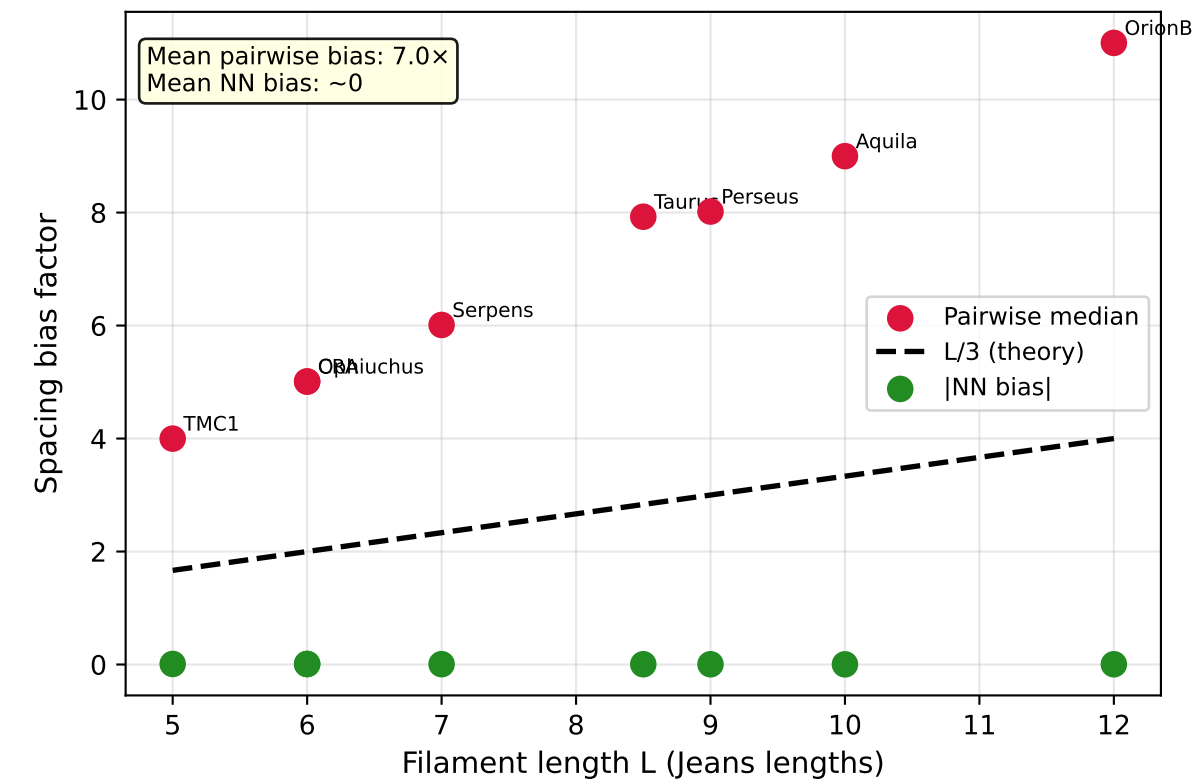
(a) C8: Fragmentation timescale vs field angle
 $f=1.5$, $\beta=0.3-2.0$, 5 seeds; $n=169$ FRAG



(b) C8_lw: λ/W calibration matrix
 $F=\text{FLAT}$ (radial collapse)



(c) C10: L/3 bias in pairwise spacing estimator
(NN estimator is unbiased: bias ≈ 0)



(d) C12: DTC $\beta=0.3$, $M=1$ stable ridge retracted
299/300 FRAG at $t_{\text{frag}} = 1.2-1.5 t_J$

